

gemini-cli

Troubleshooting guide

This guide provides solutions to common issues and debugging tips, including topics on:

- Authentication or login errors
- Frequently asked questions (FAQs)
- Debugging tips
- Existing GitHub Issues similar to yours or creating new Issues

Authentication or login errors

- **Error: Failed to login. Message: Request contains an invalid argument**
 - Users with Google Workspace accounts or Google Cloud accounts associated with their Gmail accounts may not be able to activate the free tier of the Google Code Assist plan.
 - For Google Cloud accounts, you can work around this by setting `GOOGLE_CLOUD_PROJECT` to your project ID.
 - Alternatively, you can obtain the Gemini API key from [Google AI Studio](#), which also includes a separate free tier.
- **Error: UNABLE_TO_GET_ISSUER_CERT_LOCALLY OR unable to get local issuer certificate**
 - **Cause:** You may be on a corporate network with a firewall that intercepts and inspects SSL/TLS traffic. This often requires a custom root CA certificate to be trusted by Node.js.
 - **Solution:** Set the `NODE_EXTRA_CA_CERTS` environment variable to the absolute path of your corporate root CA certificate file.
 - Example: `export NODE_EXTRA_CA_CERTS=/path/to/your/corporate-ca.crt`

Common error messages and solutions

- **Error: EADDRINUSE (Address already in use) when starting an MCP server.**
 - **Cause:** Another process is already using the port that the MCP server is trying to bind to.
 - **Solution:** Either stop the other process that is using the port or configure the MCP server to use a different port.
- **Error: Command not found (when attempting to run Gemini CLI with `gemini`).**
 - **Cause:** Gemini CLI is not correctly installed or it is not in your system's `PATH`.

- **Solution:** The update depends on how you installed Gemini CLI:
 - If you installed `gemini` globally, check that your `npm` global binary directory is in your `PATH`. You can update Gemini CLI using the command `npm install -g @google/gemini-cli@latest`.
 - If you are running `gemini` from source, ensure you are using the correct command to invoke it (e.g., `node packages/cli/dist/index.js ...`). To update Gemini CLI, pull the latest changes from the repository, and then rebuild using the command `npm run build`.
- **Error: `MODULE_NOT_FOUND` or import errors.**
 - **Cause:** Dependencies are not installed correctly, or the project hasn't been built.
 - **Solution:**
 - a. Run `npm install` to ensure all dependencies are present.
 - b. Run `npm run build` to compile the project.
 - c. Verify that the build completed successfully with `npm run start`.
- **Error: "Operation not permitted", "Permission denied", or similar.**
 - **Cause:** When sandboxing is enabled, Gemini CLI may attempt operations that are restricted by your sandbox configuration, such as writing outside the project directory or system temp directory.
 - **Solution:** Refer to the [Configuration: Sandboxing](#) documentation for more information, including how to customize your sandbox configuration.
- **Gemini CLI is not running in interactive mode in "CI" environments**
 - **Issue:** The Gemini CLI does not enter interactive mode (no prompt appears) if an environment variable starting with `CI_` (e.g., `CI_TOKEN`) is set. This is because the `is-in-ci` package, used by the underlying UI framework, detects these variables and assumes a non-interactive CI environment.
 - **Cause:** The `is-in-ci` package checks for the presence of `CI`, `CONTINUOUS_INTEGRATION`, or any environment variable with a `CI_` prefix. When any of these are found, it signals that the environment is non-interactive, which prevents the Gemini CLI from starting in its interactive mode.
 - **Solution:** If the `CI_` prefixed variable is not needed for the CLI to function, you can temporarily unset it for the command. e.g., `env -u CI_TOKEN gemini`
- **DEBUG mode not working from project `.env` file**
 - **Issue:** Setting `DEBUG=true` in a project's `.env` file doesn't enable debug mode for `gemini-cli`.
 - **Cause:** The `DEBUG` and `DEBUG_MODE` variables are automatically excluded from project `.env` files to prevent interference with `gemini-cli` behavior.
 - **Solution:** Use a `.gemini/.env` file instead, or configure the `advanced.excludedEnvVars` setting in your `settings.json` to exclude fewer variables.

Exit Codes

The Gemini CLI uses specific exit codes to indicate the reason for termination. This is especially useful for scripting and automation.

Exit Code	Error Type	Description
41	FatalAuthenticationError	An error occurred during the authentication process.
42	FatalInputError	Invalid or missing input was provided to the CLI. (non-interactive mode only)
44	FatalSandboxError	An error occurred with the sandboxing environment (e.g., Docker, Podman, or Seatbelt).
52	FatalConfigError	A configuration file (<code>settings.json</code>) is invalid or contains errors.
53	FatalTurnLimitedError	The maximum number of conversational turns for the session was reached. (non-interactive mode only)

Debugging Tips

- **CLI debugging:**
 - Use the `--verbose` flag (if available) with CLI commands for more detailed output.
 - Check the CLI logs, often found in a user-specific configuration or cache directory.
- **Core debugging:**
 - Check the server console output for error messages or stack traces.
 - Increase log verbosity if configurable.
 - Use Node.js debugging tools (e.g., `node --inspect`) if you need to step through server-side code.
- **Tool issues:**
 - If a specific tool is failing, try to isolate the issue by running the simplest possible version of the command or operation the tool performs.
 - For `run_shell_command` , check that the command works directly in your shell first.
 - For *file system tools*, verify that paths are correct and check the permissions.
- **Pre-flight checks:**
 - Always run `npm run preflight` before committing code. This can catch many common issues related to formatting, linting, and type errors.

Existing GitHub Issues similar to yours or creating new Issues

If you encounter an issue that was not covered here in this *Troubleshooting guide*, consider searching the Gemini CLI [Issue tracker on GitHub](#). If you can't find an issue similar to yours, consider creating a new GitHub Issue with a detailed description. Pull requests are also welcome!

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